🔭 Skywarden

A console logbook for astronomical observations.
Record, search, analyse and chart your nights under the sky, straight from the terminal.

 Python
 SQLite
 Matplotlib
 Status

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✨ Overview

Skywarden is a small, dependency-light program that keeps track of what you observed, with which filter, for how long, and what you measured.
It supports two kinds of observations, built with simple object-oriented classes, and can export your data to **CSV** or **SQLite** and plot exposure times as a **bar chart**.

```
========================================

          Welcome to Observer Log

========================================
 1. Add Photometric Observation
 2. Add Spectroscopic Observation
 3. Show Observation
 4. Show Average exposure
 5. Show Maximum exposure
 6. Show Minimum exposure
 7. Show Total exposure
 8. Search target
 9. Save to a csv file
10. Save to a database
11. Show bar chart for target vs exposure time
12. Exit
```

🌌 Features

	Feature	Description
🌟	**Photometric observations**	Target, observer, filter, exposure time and magnitude
🌈	**Spectroscopic observations**	Target, observer, filter, exposure time and wavelength (nm)
🛡️	**Input validation**	Rejects letters, `nan`, `inf`, zero and negative values where they make no sense
📊	**Exposure statistics**	Average, maximum, minimum and total exposure time
🔎	**Search**	Find observations by target name, case-insensitive
💾	**CSV export**	One file with magnitude and wavelength columns
🗄️	**SQLite storage**	Save the session to `observation.db`
📈	**Bar chart**	Target vs exposure time with Matplotlib

🧬 How it fits together

```
classDiagram
    class Observation {
        +target
        +observer
        +filter_name
        +exposure_seconds
        +summary()
        +exposure_minutes()
    }
    class PhotometricObservation {
        +magnitude
        +summary()
    }
    class SpectroscopicObservation {
        +wavelength_nm
        +summary()
    }
    Observation <|-- PhotometricObservation
    Observation <|-- SpectroscopicObservation
```

🗂️ Project structure

```
Skywarden/
├── main.py            # menu, input handling, statistics, search, CSV export
├── observation.py     # Observation, PhotometricObservation,
SpectroscopicObservation
├── database.py        # SQLite saving
├── visualization.py   # Matplotlib bar chart
└── README.md
```

🚀 Getting started

1. Clone the repository

```
git clone https://github.com/amirmohammadarefkhani1385-creator/Skywarden.git
cd Skywarden
```

2. Install the only external dependency

```
pip install matplotlib
```

3. Run

```
python main.py
```

> Python 3.10 or newer is recommended. `csv`, `math` and `sqlite3` ship with Python.

🛰️ Example session

```
Choose: 1
Target: Vega
Name of observer: Amir
Filter name: V
Exposure time (second): 120
Magnitude: 0.03
Photometric observation added Succesfully

Choose: 3

Observation >>> 1
  Target   : Vega
  Observer : Amir
  Filter   : V
  Exposure : 120 s (2.00 min)
  Magnitude: 0.03
```

💾 Where your data goes

Option	Output	Contents
`9` CSV	`<your name>.csv`	target, observer, filter, exposure, magnitude, wavelength
`10` Database	`observation.db`	table `observations`: target, observer, filter, exposure
`11` Chart	window	bar chart of exposure time per target

🧰 Built with

- **Python**: core logic and classes
- **Matplotlib**: visualisation
- **SQLite**: local database
- **csv / math**: file export and number validation

👤 Author

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Clear skies ✨